

Illustrative Case Study (Land-Use Change DDABM)

To further demonstrate vODD-DD beyond the epidemiology domain, we present a second illustrative case: a residential land-use change DDABM inspired by Parker et al. (2003) and Brown et al. (2005), which has become the canonical framework for spatially explicit, data-driven simulation of urban land-use dynamics. In this model, we assume that households and developers act as autonomous agents over a parcel raster, household preferences are estimated from observed transaction and demographic data, and the model is calibrated against historical land-use maps via pattern-oriented modeling. This case provides clearly identifiable agents (households, developers), interaction rules (parcel bidding, neighbor-based amenity effects), and a structured environment (zoned parcel grid), which align well with the requirements of the proposed protocol.

The purpose of the model is to simulate residential location choice and developer conversion to evaluate the impact of land-use policies: zoning, urban-growth boundaries (UGB), and transit-oriented development (TOD) upzoning, and to support Digital Twin decision-making for regional planning. The model represents a metropolitan region as a regular raster of $100\text{ m} \times 100\text{ m}$ parcels (cell state: undeveloped U, developed D; layers: zoning, accessibility, amenity, existing development), populated by N household agents (attributes: income, preference class, tenure status) and a small set of developer agents, with N scaled to the regional total. At each annual step, dissatisfied households evaluate feasible parcels and submit utility-based bids (multinomial logit over price, accessibility, and amenity); the highest bid wins. Developers convert $U \rightarrow D$ when expected return exceeds a threshold τ . Neighbor-based externalities update local aesthetic and price within a Moore neighborhood each step. Policy interventions such as a strict growth boundary, TOD upzoning, and their combination are encoded as constraints on the feasible-parcel set and as density-allowance multipliers, with calibrated preference weights drawn from posterior distributions.

Table A summarizes the model components, key assumptions, and data sources; Table B summarizes the data-driven workflow from collection through calibration, validation, and scenarios.

Table A: Land-use DDABM components, assumptions, and data sources.

ABM Component	Sub-category	Element	Value / Assumption	Data Source
Agents	State	Land-use state per parcel	U (undeveloped), D (developed)	—
	Attributes	Demographics	Income, household composition, tenure status	Census microdata (static)
	Population	N (household agents)	Scaled to regional total; developer agents added	Census microdata (static)
	Behaviors	Move decision	Triggered by life-cycle events or dissatisfaction > threshold	Real-estate transactions (dynamic)
	Behaviors	Location choice	Utility-based bid (multinomial logit) over feasible parcels	Real-estate transactions (dynamic)
	Behaviors	Developer conversion	$U \rightarrow D$ if expected return > τ	Real-estate transactions (dynamic)
Interaction Rules	Bidding	Parcel competition	Highest bid wins; ties broken stochastically	Real-estate transactions (dynamic)
	Neighbor effects	Amenity update	Local density & land cover update aesthetic and price within Moore neighborhood	Historical land-use maps (dynamic)
	Agent–Environment	Feasibility	Zoning and growth-boundary layers constrain the feasible-parcel set	Zoning & policy layers (dynamic)
Environment	Grid	Spatial structure	Regular raster, 100 m × 100 m cells	Parcel cadastre (static)
	Layers	Zoning	Allowed uses and density caps per cell	Zoning & policy layers (dynamic)
	Layers	Accessibility	Network distance to CBD and transit	Road & transit GIS (static/dynamic)
	Layers	Amenity	Lakes, parks, views, neighborhood quality	GIS feature layers + hedonic regression (static)
Other Key Parameters				
Interventions	Policies	UGB, TOD upzoning, combined	Spatial constraint on feasible parcels; density-allowance multipliers near transit	Zoning & policy layers (dynamic)
Preference weights, τ , neighbor radius	Behavioral parameters	Parameters	Estimated via calibration	Real-estate + Historical land-use maps
Time step	Temporal resolution	Step size	1 year	—
Duration / Stop	Simulation length	End condition	40 years (1990–2030); stop at horizon or when developable land exhausted	—

Table B: Data-driven workflow summary for the land-use DDABM.

Stage	Details	Data / Input	Method	Output / Role
Data Inputs & Pipeline	Collection	Parcel cadastre, census microdata, NLCD/Landsat land-use maps, MLS transactions, zoning layers	County GIS export, USGS NLCD archive, census API, MLS/assessor feed	Raw datasets for initialization & calibration
	Pre-processing	All collected datasets	Parcel-to-raster rasterization (100 m); land-cover reclassification (U/D); spatial join of demographics to parcels	Cleaned 100 m grid with parcel attributes
	Analysis	Pre-processed data	Hedonic regression for amenity weights; k-means clustering of move patterns → preference classes; accessibility surface (network distance)	Initial amenity-weight estimates, preference-class assignments, accessibility layer
Model Evaluation	Calibration	Historical land-use maps (1990 → 2010 trajectory)	Pattern-oriented modeling (POM) with genetic algorithm over preference weights, developer threshold τ , neighbor-effect radius	Best-fit parameter set; multi-resolution Kappa > 0.70 over calibration period
	Output validation	Held-out 2010–2020 land-use maps	Multi-resolution Kappa; density-gradient and fractal-dimension comparison	Quantified spatial accuracy on unseen period
	Scenario validation	Post-policy implementations (e.g., adopted growth boundaries)	Simulated vs recorded post-policy land-use trends	Confirmed model responsiveness to policy levers
	Null-model contrast	Calibrated model vs random allocation and persistence null models	Pattern-metric comparison against null baselines	Demonstrates calibrated model clearly outperforms naive baselines
Scenarios (Simulation Experiments)	Baseline	Calibrated preferences; status-quo zoning	Standard land-use ABM run	Reference settlement trajectory
	Growth boundary	Spatial constraint around existing built-up area	Feasible-parcel set restricted to within UGB	Compacted development; potential leapfrog at boundary
	TOD upzoning	Density-allowance multiplier within 800 m of transit corridors	Modified zoning layer near transit	Polycentric structure; corridor densification
	Combined policies	UGB + TOD upzoning	Joint application of both constraints	Key scenario for DT-informed policy recommendation

References

- Parker, D.C., Manson, S.M., Janssen, M.A., Hoffmann, M.J., & Deadman, P. (2003). *Multi-agent systems for the simulation of land-use and land-cover change: A review*. **Annals of the Association of American Geographers**, 93(2), 314–337.
- Brown, D.G., Page, S., Riolo, R., Zellner, M., & Rand, W. (2005). *Path dependence and the validation of agent-based spatial models of land use*. **International Journal of Geographical Information Science**, 19(2), 153–174.